

10/12/2025

Designing a DAB using Scaling Laws

Reference:

https://ridleyengineering.com/images/webinar/DAB_Design_Using_Scaling_Laws.pdf

Aims:

LV stage:

$V_{nom} = 48 \text{ V}$

$V_{min} = 24 \text{ V}$

$V_{max} = 60 \text{ V}$

$P_{nominal} = 2 \text{ kW}$

$I_{max} = \sim 42 \text{ A}$

HV stage:

$V_{nom} = 400 \text{ V}$

$V_{max} = 420 \text{ V}$

$V_{min} = 360 \text{ V}$

$P_{nominal} = 2 \text{ kW}$

$I_{max} = 5 \text{ A}$

Fsw starting point = 100 kHz

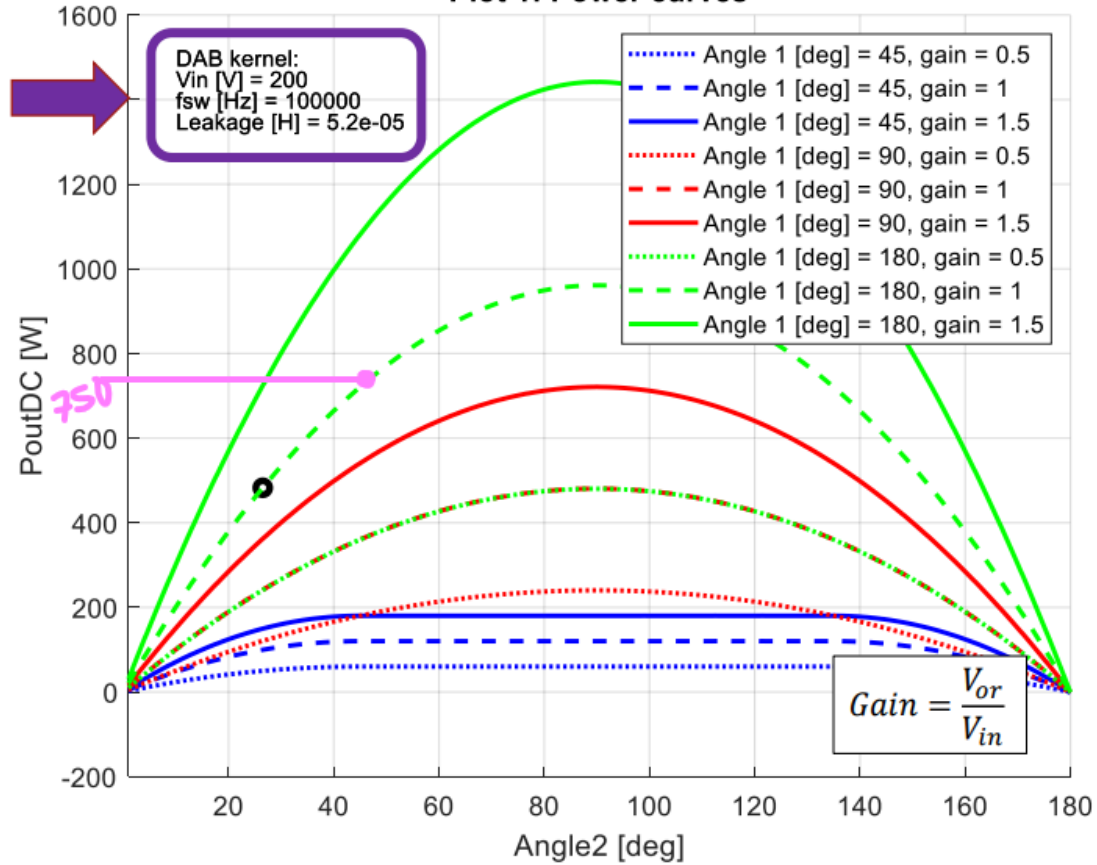
Choose a Single Phase Shift mode

I.e. (Intra bridge) Angle 1 = 180°

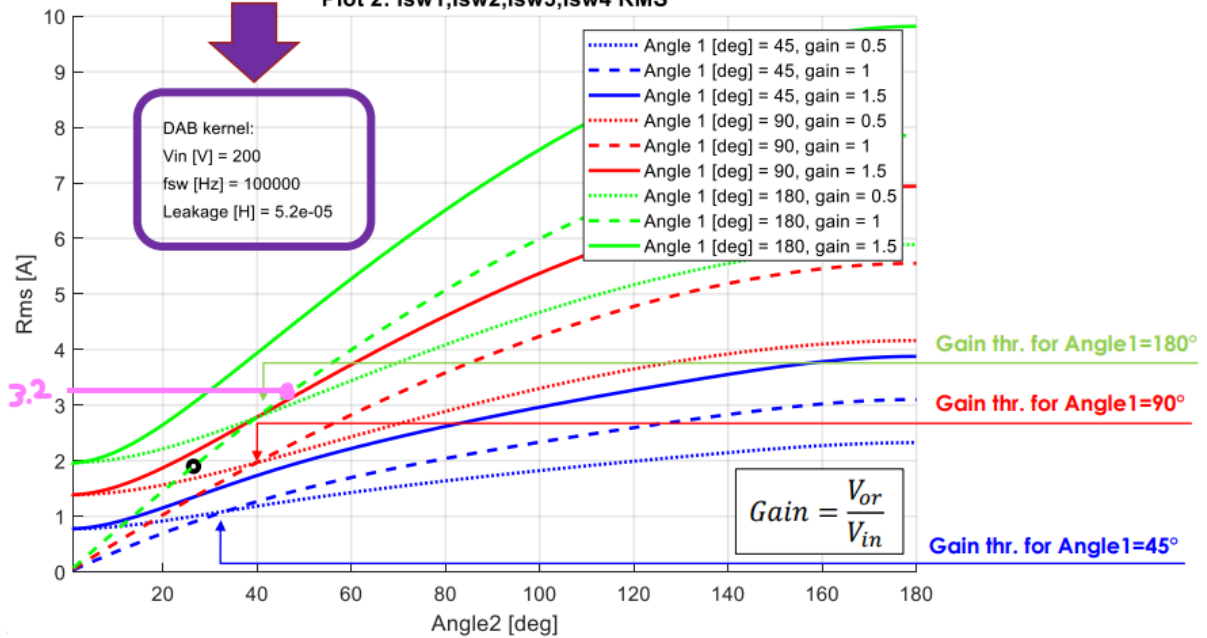
(Inter bridge) Angle 2 = Control angle up to 45°

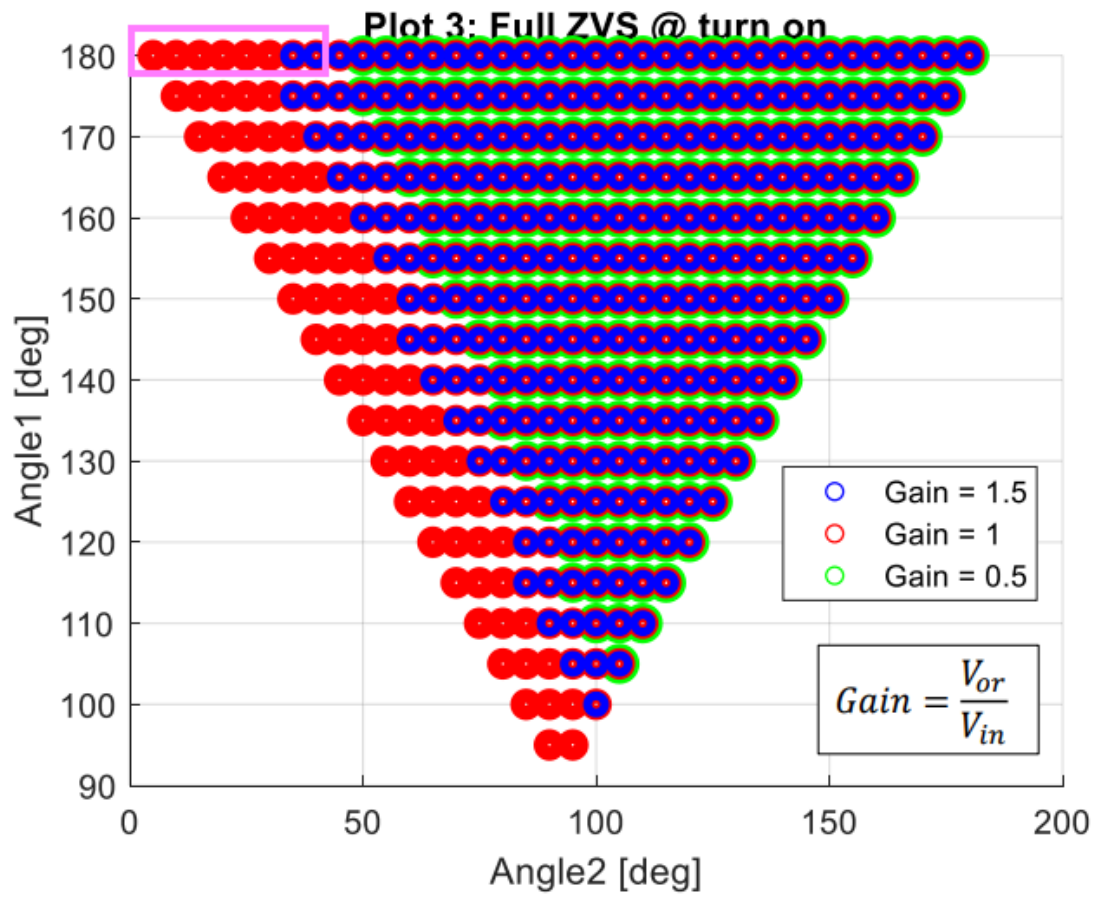
Gain = 1

Plot 1: Power curves



Plot 2: Isw1, Isw2, Isw3, Isw4 RMS





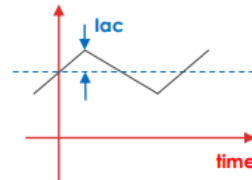
The Scaling laws: « three is a magic number! »

- To double the power at a certain frequency (keeping all else unchanged such as the phase angles), we need to halve the inductance and double the capacitance.
- To double the frequency, keeping the same power, once again we just need to halve the inductance and halve the capacitance. Recalling the buck converter equations.

$$L = \frac{V_{out} \cdot (1-D)}{r \cdot f_{sw} \cdot I_{out}}$$

$$C = \frac{I_{out} \cdot r}{8 \cdot f_{sw} \cdot \Delta V}$$

$$r = \frac{2 \cdot I_{ac}}{I_{dc}}$$



- Added guest to the party: If we double the input voltage, we will quadruple the power. Proof below:**

If:

$$\triangleright P_{out} \cong P_{in}$$

$$\triangleright V_{out} = TF_{dc} \cdot V_{in}$$

then:

$$\triangleright P_{out} = \frac{V_{out}^2}{R_{load}} = \frac{(TF_{dc} \cdot V_{in})^2}{R_{load}}$$

If Vin doubles, to keep the equation valid, the power shall quadruple!

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DAB Design 1

Pout = 7 kW, Vout = 48V, Vin = 380V, fsw = 100kHz

Assumptions

- Best entry Angles are Angle 1 = 180° and Angle 2 ≈ 26°
- Best gain entry point is: gain = 1

Before: calculate the scaling factors!

$$k_{voltage} = \frac{V_{in\,eff}}{V_{kernel}} = \frac{380}{200} = 1,9 \rightarrow \text{Power boost} = 1,9^2$$

$$k_{power} = \frac{P_{eff}}{P_{kernel}} = \frac{7k}{480 \cdot 1,9^2} = 4$$

$$k_{frequency} = \frac{f_{eff}}{f_{kernel}} = \frac{100k}{100k} = 1$$

Scaling from
the kernel to
the new
requirements

One Page
DAB design

After: calculate the Leakage inductance, the turn ratio and both capacitors):

$$L_{Lk} = \frac{L_{kernel}}{k_f \cdot k_p} = \frac{52\,uH}{4} = 12,8\,uH$$

$$C_{dc\,block} \approx 100 \cdot \frac{1}{4 \cdot \pi^2 \cdot f^2 \cdot L_{lk}} \approx 20\mu$$

$$N = \frac{N_p}{N_s} = \frac{380}{48} = 7.91$$

$$C_{out} \approx \left(I_{Llk\,peak} \cdot \frac{N_p}{N_s} - \frac{P_{out}}{V_{out}} \right) \cdot \frac{T}{\Delta V_{out\,(1V)}} \approx 110\mu$$

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Kernel power: 750 W (Angle 1 = 180 °, Angle 2 = 45 °, Gain = 1)

Kernel Fsw = 100 kHz

Kernel Leakage = 5.2e-05 H

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Kernel Input voltage: 200 V

$I_{Llk_peak} = 10 \text{ A}$

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$V_{min} = 360 \text{ V}$

$P_{nominal} = 2 \text{ kW}$

$I_{max} = 5 \text{ A}$

Fsw starting point = 100 kHz

$T = 1 / 100 \text{ kHz} = 10 \text{ us}$

$T_{angle2} = 1.25 \text{ us}$

Choose a Single Phase Shift mode

I.e. (Intra bridge) Angle 1 = 180°

(Inter bridge) Angle 2 = Control angle up to 45°

Gain = 1

$k_{voltage} = V_{in_eff} / V_{kernel} = 48 / 200 = 0.24 \rightarrow \text{Power "boost"} 0.24^2 = 0.058$

$k_{power} = P_{eff} / P_{kernel} = 2000 / (750 * 0.24^2) = 46.3$

$k_{frequency} = f_{eff} / f_{kernel} = 100 \text{ k} / 100 \text{ k} = 1$

$L_{lk} = L_{kernel} / (k_{frequency} \times k_{power}) = 52 \text{ uH} / (1 * 46.3) = 1.12 \text{ uH}$

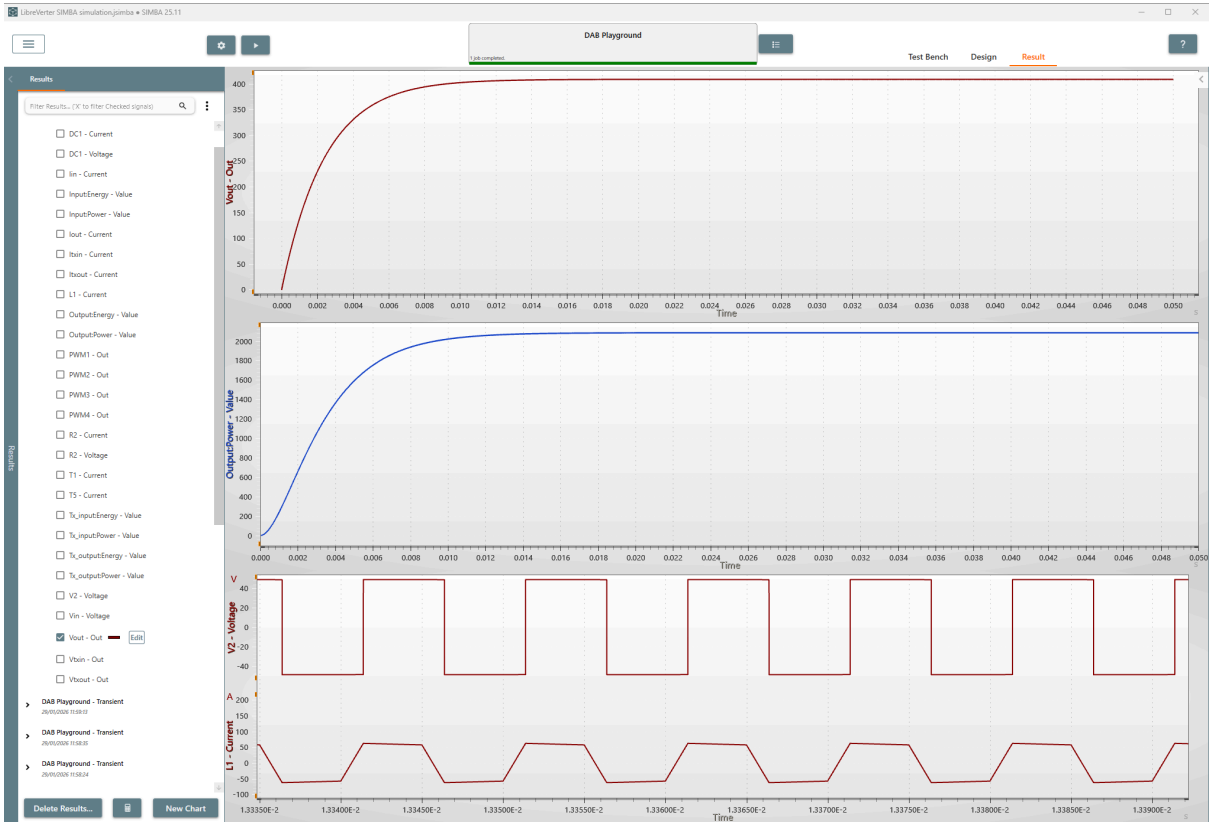
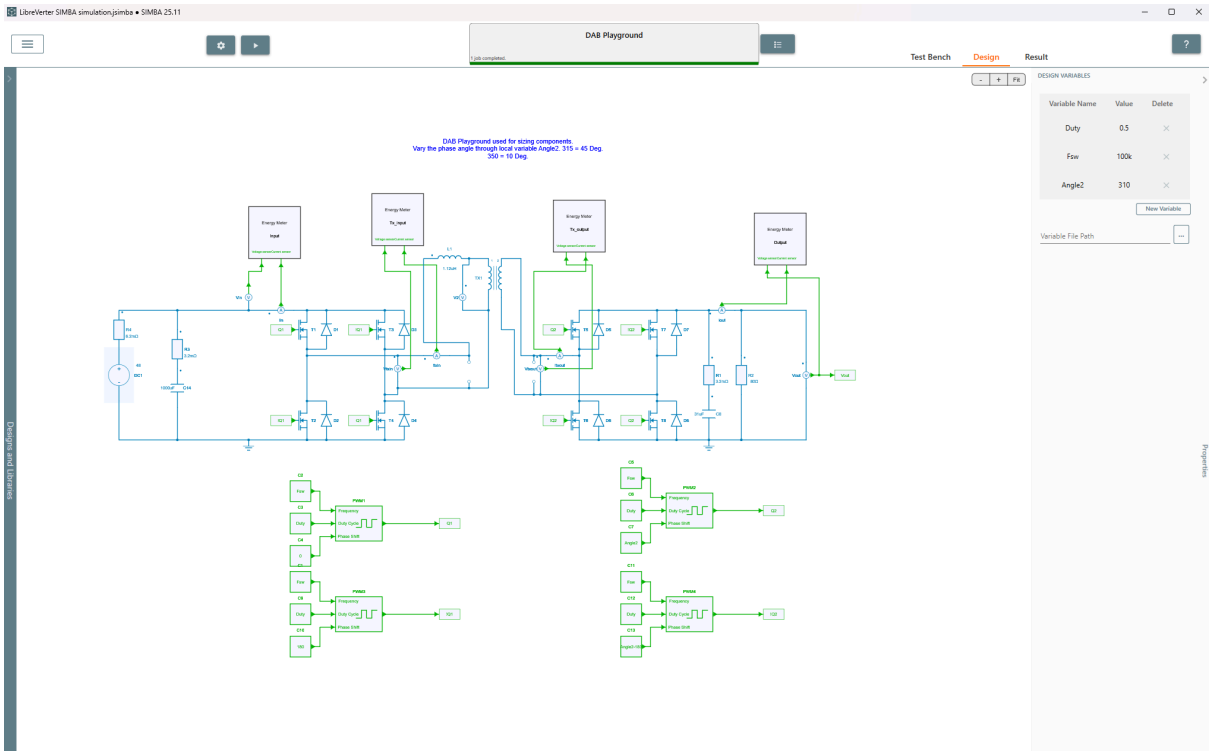
$C_{dcblock} = 100 \times (1 / (4 \times \pi^2 \times f^2 \times L_{lk})) = 226 \text{ uF}$

$N = N_p / N_s = 48 / 400 = 0.12$

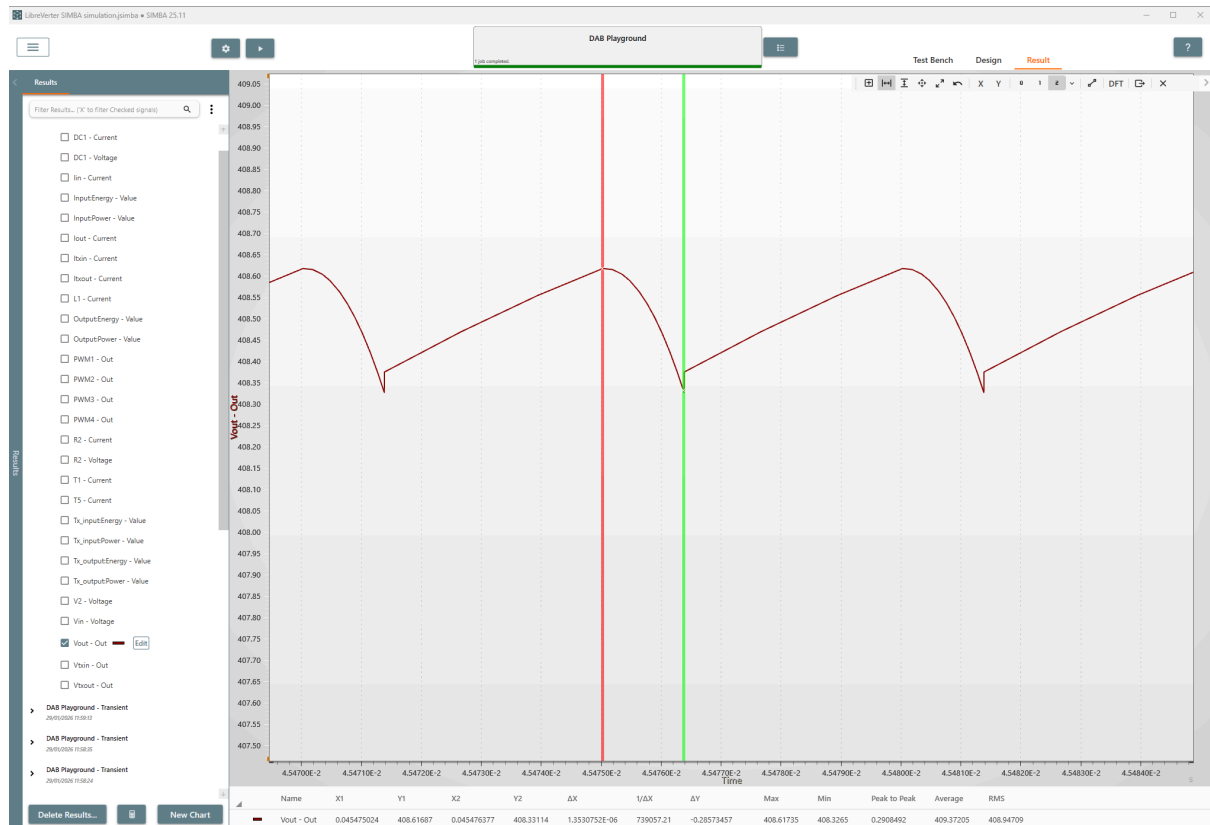
$I_{Llk_peak} = (I_{Llk_peak_kernel} \times k_{voltage}) \times (P_{eff} / P_{kernel}) = (10 \times 0.24) \times (2000 / 750) = 111.11 \text{ A}$

$C_{out} = (I_{Llk_peak} \times N - P_{out} / V_{out}) \times (T/2 - T_{angle2}) / \Delta V_{out} = ((111.11 \times 0.12) - 2000/400) \times ((10\text{u}/2 - 1.25\text{u}) / 1) = 31.25 \text{ uF}$

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Validation:

408 V on 80 Ohm resistor $\rightarrow P = V^2 / R = (408^2) / 80 = 2080.8 \text{ W}$

Voltage ripple $\sim 0.3 \text{ V}$

Input into openmagnetics:

Core: <https://uk.rs-online.com/web/p/products/6479373/>

Former: <https://uk.rs-online.com/web/p/products/2133922P/>

Yoke: <https://uk.rs-online.com/web/p/products/6479418/>

Windings: <https://www.rmcybernetics.com/shop/wire-cable/litz-wire>

“12A” (0.05 mm x 1000) and “3A” (0.1 mm x 75) (variants for primary and secondary respectively. Should be possible to wind to get $\sim 450 \text{ nH}$ leakage

See OpenMagnetics output file!